

OUTPUT

Main Menu

```
=====
SMART CAMPUS INFORMATION SYSTEM
=====
1. Register Student
2. Enroll Courses
3. Display Records
4. Sort And Search
5. Fee Menu
6. File Handling
7. Scan Directory
8. Performance Analysis
9. Exit
=====
Choice:
```

Scenario 1: Selecting Option 1

```
Enter your choice: 1
Enter student name: Tanu
Enter student ID: 2
Enter marks (0-100): 87

===== STUDENT REPORT =====
Name      : Tanu
ID         : 2
Marks      : 87
Grade      : A+
Remark     : Excellent
```

```
Enter your choice: 1
Enter student name: Shreya
Enter student ID: 1
Enter marks (0-100): 78

===== STUDENT REPORT =====
Name      : Shreya
ID         : 1
Marks      : 78
Grade      : A
Remark     : Very Good
```

Scenario 2: Selecting Option 2

```
Enter your choice: 2
Enter student ID: 2

Add maximum 5 courses
Type 'done' to stop

Enter course name: Maths
Enter course name: done
Courses enrolled successfully!
```

Scenario 3: Selecting Option 3

```
===== STUDENT RECORDS =====
ID      : 1
Name    : Shreya
Marks   : 78
Grade   : A
Courses : Maths
-----
ID      : 2
Name    : Tanu
Marks   : 87
Grade   : A+
Courses : Maths
-----
```

Scenario 4: Selecting Option 4

```
Sorted Student IDs:
['1', '2']

Enter ID to search: 2
Student ID found at position 1
```

Scenario 5: Selecting Option 5

```
Enter hostel fee (0 if none): 50000
Enter transport fee (0 if none): 0

===== FEE DETAILS =====
Tuition Fee : 50000
Hostel Fee   : 50000
Transport    : 0
Total Fee    : 100000
```

Scenario 6: Selecting Option 6

```
Records written successfully!

===== FILE CONTENT =====
ID,Name,Marks,Grade
1,Shreya,78,A
2,Tanu,87,A+
```

Scenario 7: Selecting Option 7

```
Enter directory path: C:\Users\Anushka\OneDrive\Desktop\New folder
Scanning Directory...

New folder/
  22ESC142- Module -3 - DC Generator_motor_notes SJ and SA.pdf
  2nd module python.docx
  chem mod 2.pdf
  chem mod 3 .pdf
  mod 3python.docx
```

Scenario 8: Selecting Option 8

```
===== STUDENT PERFORMANCE DATA =====
```

	Name	Marks
0	Shreya	78
1	Tanu	87

```
===== STATISTICAL SUMMARY =====
```

	Marks
count	2.000000
mean	82.500000
std	6.363961
min	78.000000
25%	80.250000
50%	82.500000
75%	84.750000
max	87.000000

Average Marks : 82.5

Top Performer : Tanu

Top Marks : 87

